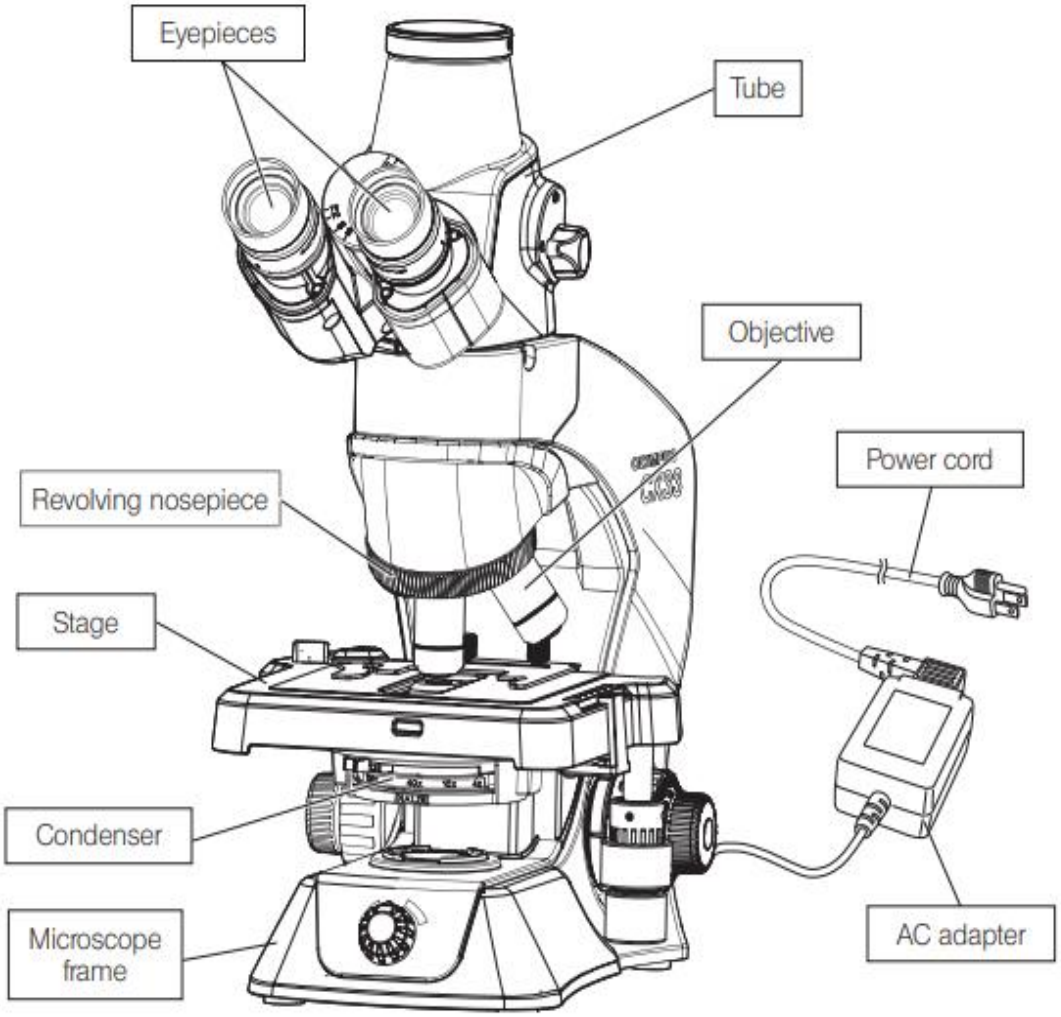
Biologik mikroskop

Foydalanish bo'yicha qo'llanma



Ishlab chiqaruvchi	**Olympus**
Mahsulot nomi (Cat. No)	**Biologik mikroskop (CX33)**

I . Standart kombinatsiya



Option units

- Eyepiece (2 pieces) WHSZ15X-H



- Filter holder CH2-FH



- Darkfield ring CH2-DS



- 20X objective
  100X objective
- Eyepiece micrometer (Ø24mm, Thickness: 1.5mm)
- Ø45mm filter
- Camera adapter U-TV1xC



- Camera DP22 or DP27



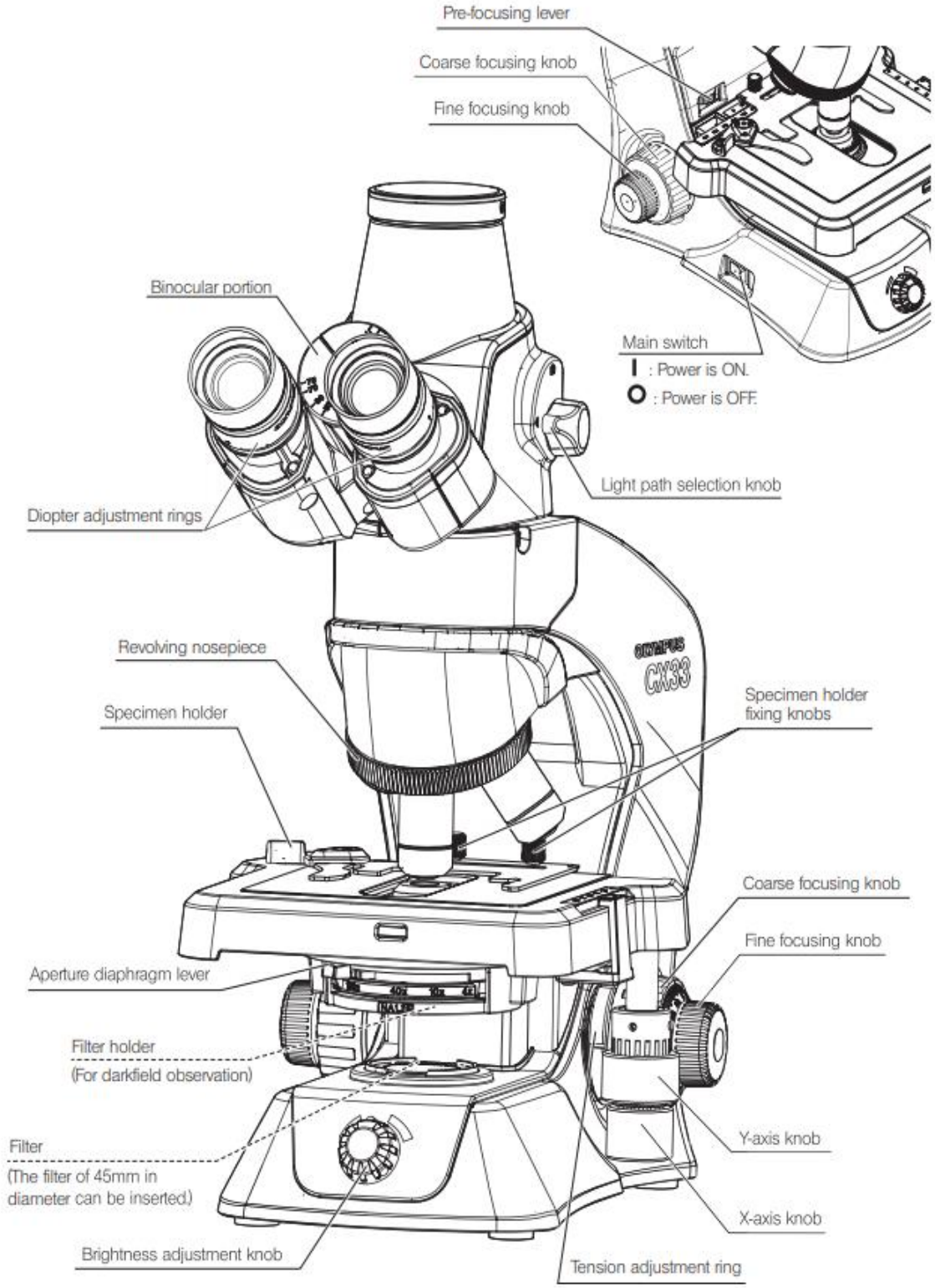
- Specimen hold plate CX3-SHP



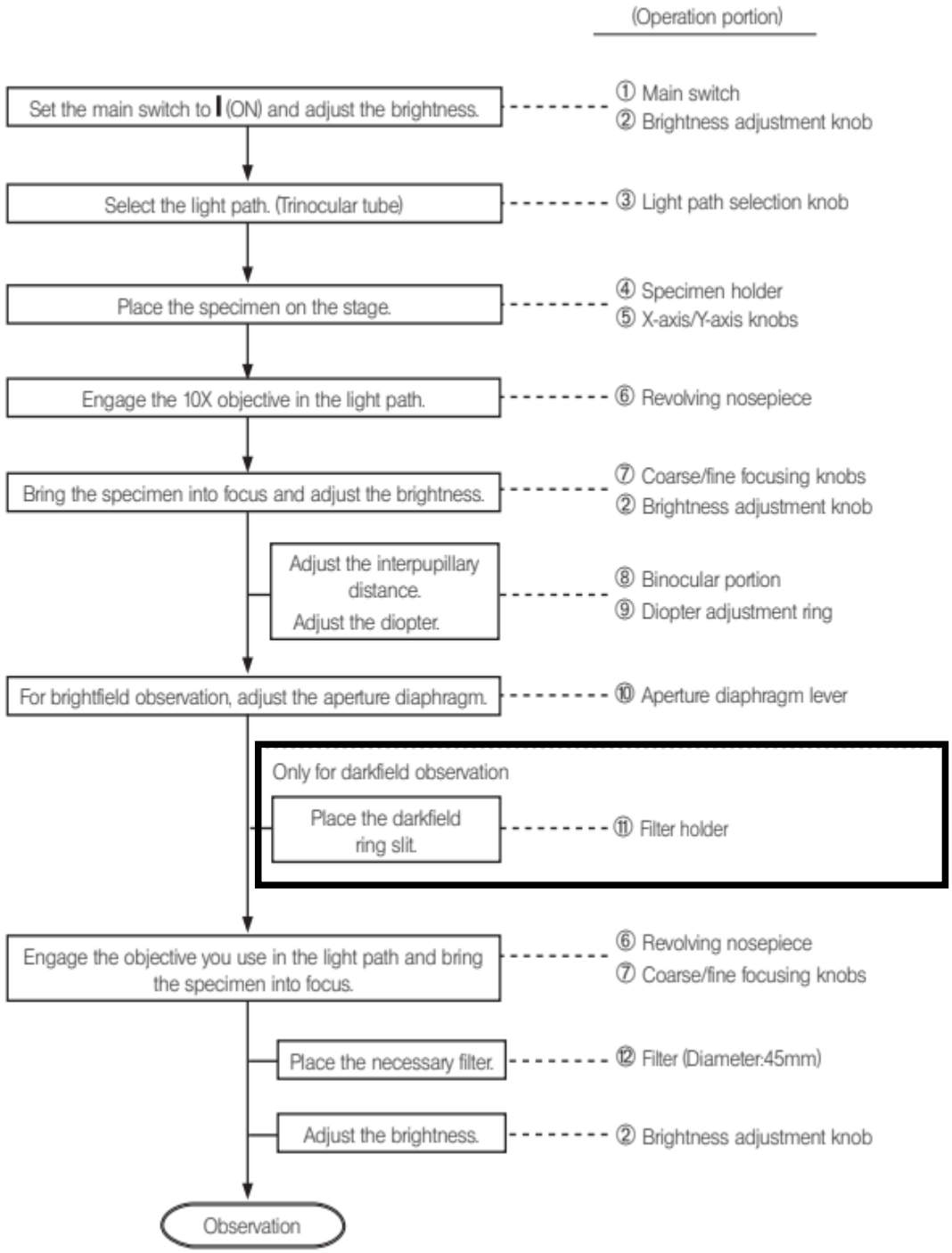
- Specimen holder CX3-HLDT

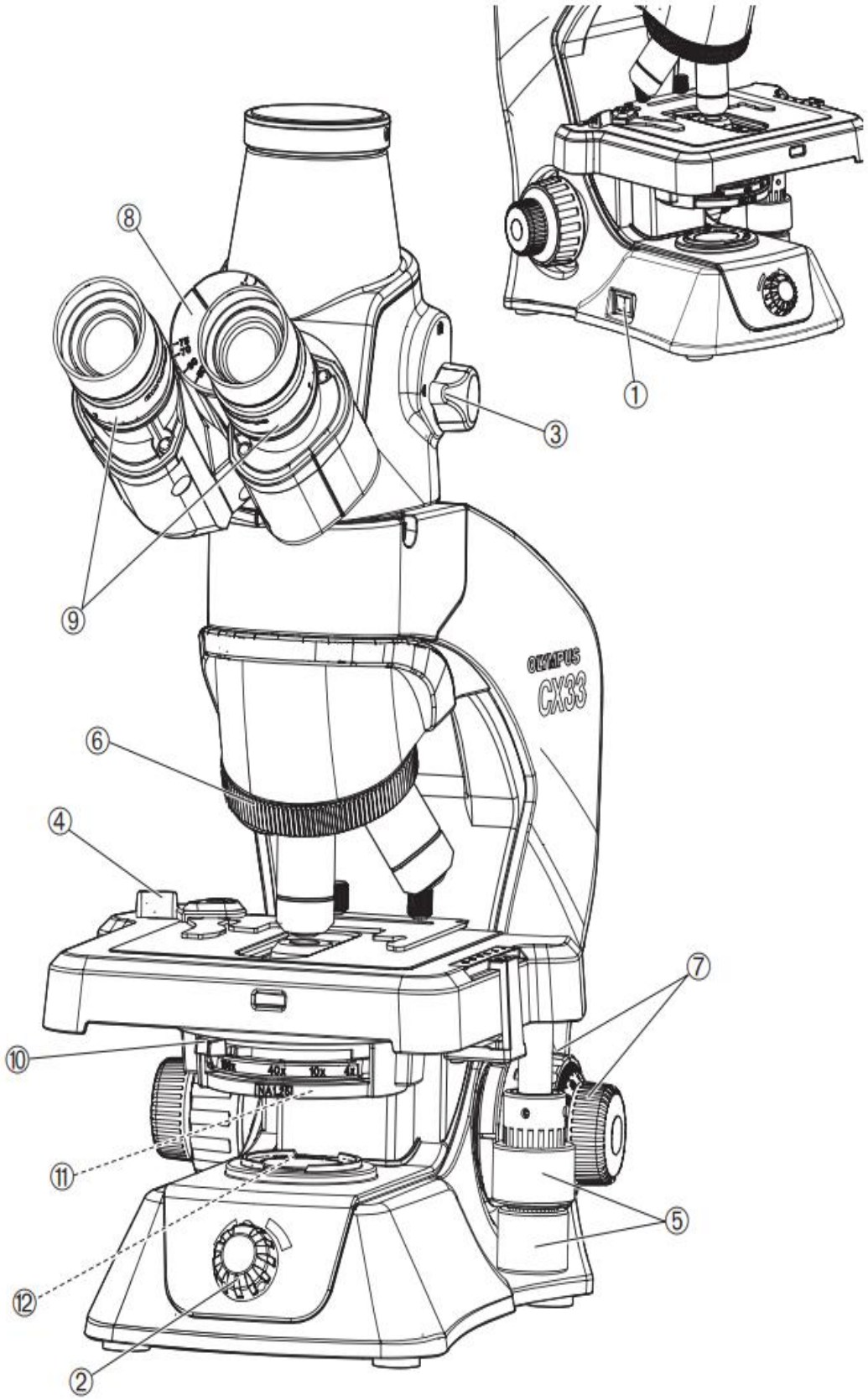


Ⅱ. Boshqaruv qismlari nomlanishi



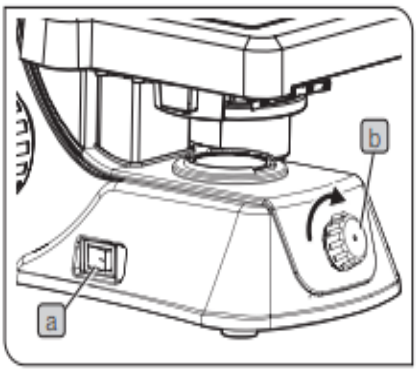
Ⅲ. Yorug‘ maydon va qorong‘i maydon kuzatuv usullarining umumiy tavsifi




OLYMPUS
CX33
40x
10x
4x
NA1.25
①
②
③
④
⑤
⑥
⑦
⑧
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IV. Kuzatuv tartiblari

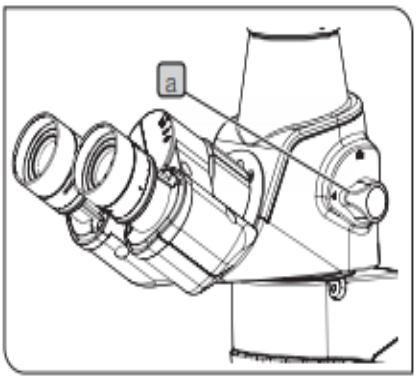
1. LED yoritgichini yoqish



1) Asosiy kalit **“a” ni ON** holatiga o‘tkazing.

2) Yorqinlikni sozlash tugmasi **“b”** ni strelka yo‘nalishida aylantirsangiz, yorqinlik oshadi, qarama-qarshi yo‘nalishda aylantirsangiz, yorqinlik kamayadi..

2. Okulyar yorug'lik yo'li va kamera yorug'lik yo'li o'rtasida tanlov



Tasvirni okulyar orqali kuzatish yoki kamera orqali monitorlarda ko‘rish uchun yorug‘lik yo‘lini tanlashingiz mumkin.

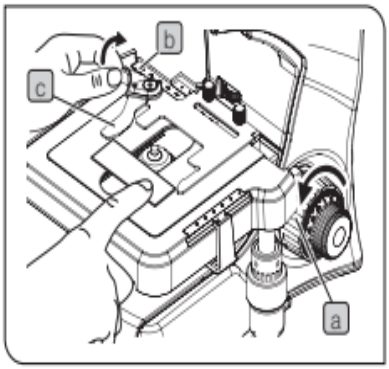
1) Trinokulyar trubkadagi yorug‘lik yo‘lini tanlash tugmasi **“a”** ni aylantirib, kerakli yo‘lni tanlang. (Yorug‘lik intensivligi nisbati: 100:0)

: Eyepiece
: Camera

ESLATMA : To'g'ri kuzatuv tasvirini olish uchun yorug'lik yo'lini tanlash tugmachasini to'liq to'xtaguncha aylantiring.

3. Namunani joylashtirish

Bitta namunani kuzatish uchun namuna ushlagichi



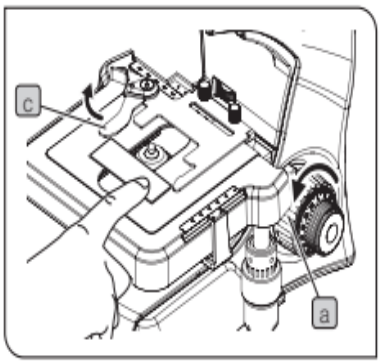
1) Makrovint(yirik) fokuslash tugmasi “a” ni o‘q yo‘nalishida aylantirib, stolchani (stage) to‘liq pastga tushiring.

2) Namuna ushlagich richagi tugmasi “b” ni orqaga (o‘q yo‘nalishida) bosing va richag “c” ni oching, so‘ng namunani old tomondan orqaga qarab stolcha ustiga surib joylashtiring.

3) Namunani joylashtirgandan so‘ng, ushlagich richagi “c” ni ehtiyotkorlik bilan qaytaring.

ESLATMA Namunani ehtiyotkorlik bilan joylashtiring. Agar richag “c” ni kuch bilan qaytarsangiz yoki “b” tugmasini o‘rtada qo‘yib yuborsangiz, richag namunaga urilib uni shikastlashi mumkin.

Namunani bir qo'li bilan joylashtirish



1) Makrovint(yirik) fokuslash tugmasi “a” ni strelka yo‘nalishida aylantirib, stolchani (stage) to‘liq pastga tushiring..

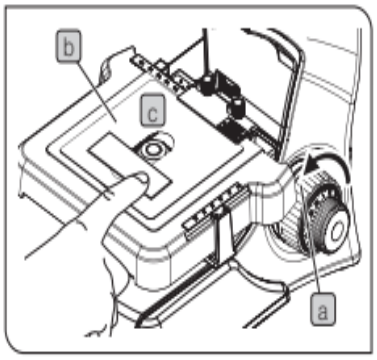
2) Namunani stolchaning old qismiga joylashtiring.

3) Namuna ushlagich richagi “c” ni rasmda ko‘rsatilganidek namunaning chetiga qo‘ying va namunani old tomondan orqaga qarab stolcha ustida surib joylashtiring.

CX3-SHP namuna ushlagich plastinasi bilan ishlash

1) Makrovint “a” ni strekla yo‘nalishida aylantirib, stolchani (stage) to‘liq pastga tushiring.

Namunani smola (resin) listi **“b”** ustiga joylashtiring**.**



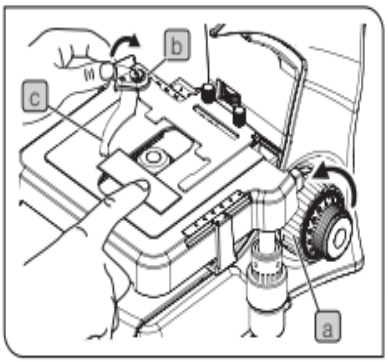
ESLATMA :

· Smola plastinkasidagi diafragma “c” tashqi qismi kondensor nuri bilan yoritilmaydi, shu sababli bu qismdagi namunani kuzatib bo‘lmaydi.
X-o‘qi tugmasini ishlatishda stolcha teshigi plastinka bilan yopilib qolmasligiga e’tibor bering**.**

· Namuna ushlagich plastinkasini 100X moyli immersion obyektiv bilan birga ishlatib bo‘lmaydi, chunki bu holatda optik xususiyatlar yetarli darajada ta’minlanmaydi.

· Namuna ushlagich plastinkasi qorong‘i maydon (darkfield) kuzatuvida ishlatilmaydi, chunki bunday kuzatuvda ko‘rish maydonining markazida soya paydo bo‘ladi.

CX3-HLDT namuna ushlagichidan foydalanganda



1) Makrovint (fokuslash) tugmasi **“a”** ni strelka yo‘nalishida aylantirib, stolchani (stage) to‘liq pastga tushiring.

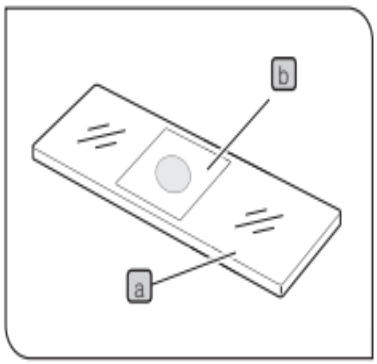
2) Namuna ushlagich richagi tugmasi **“b” ni** orqaga (strelka yo‘nalishida) bosing, richag **“c” ni** oching va namunani old tomondan orqaga qarab stolcha ustiga surib joylashtiring.

3) Birinchi namunani oxirigacha joylashtirgandan so‘ng, ikkinchi namunani birinchisiga tegadigan qilib joylashtiring.

4) Namunalarni joylashtirgandan keyin, ushlagich richagi **“c”** ni ehtiyotkorlik bilan qaytaring.

ESLATMA : Namunani ehtiyotkorlik bilan joylashtiring. Agar richag **“c”** ni kuch bilan qaytarsangiz yoki **“b”** tugmasini o‘rtada qo‘yib yuborsangiz, richag predmet oynasiga urilib uni shikastlashi mumkin.

Predmet oynasi (Slide glass)



Quyidagi **"a" slayd oynalaridan foydalanish tavsiya etiladi.**

Uzunligi: 76 mm, kengligi: 26 mm
Qalinligi 0,9 dan 1,4 mm gacha
(Qorong'u maydonni kuzatish uchun qalinligi 0,9 dan 1,2 mm gacha)

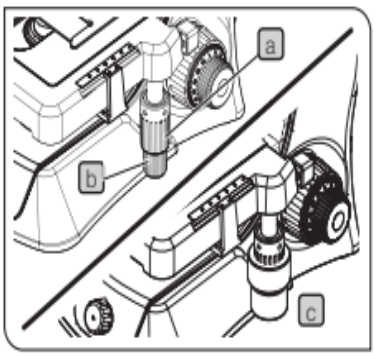
ESLATMA : Ob'ektivning to'liq ishlashini ta'minlash uchun qalinligi 0,17 mm bo'lgan **"b"** qopqoq oynasidan foydalaning .

Namunani harakatlantirish

· Yuqori Y-o‘qi tugmasi “a” ni aylantirganda, namuna Y yo‘nalishda (oldinga va orqaga) harakatlanadi.

· Pastki X-o‘qi tugmasi “b” ni aylantirganda, namuna X yo‘nalishda (o‘ngga va chapga) harakatlanadi.

Stolcha (stage) harakat diapazoni:
Chuqurlik: 52 mm × Kenglik: 76 mm



ESLATMA :

· Namunani qo‘l bilan bevosita ushlab (namuna ushlagichiga tegib) harakatlantirmang, chunki bu tugmalarning aylanish mexanizmini shikastlashi mumkin.

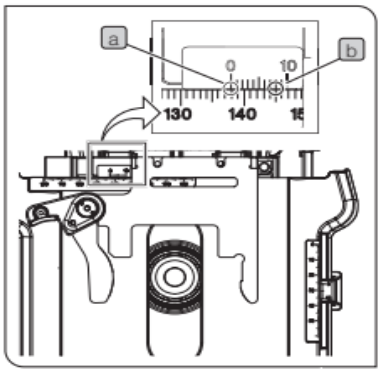
· X va Y o‘qi tugmalarini aylantirish qarshiligi (aylanish qattiqligi) harakat diapazonining oxiriga yaqinlashganda ortadi. Bunday holatda tugmani aylantirishni to‘xtating.

MASLAHAT :

· Agar stolcha tugmasidagi rezina qoplama “c” o‘rnatilgan bo‘lsa va stolchani boshqarish qiyinlashsa, ushbu rezina qoplamani olib tashlang.
Agar rezina qoplamaning balandligini o‘zgartirmoqchi bo‘lsangiz, uni yuqoriga yoki pastga siljitib sozlang.

· Stolcha tugmasi rezina qoplamasini o‘rnatishda, kattaroq qismini Y o‘qi tugmasiga (yuqori qism), kichikroq qismini esa X o‘qi tugmasiga (pastki qism) pastdan kiritib o‘rnating.

Namuna joylashuvi shkalasi (Scale of specimen position)



Namunaning kuzatuv nuqtasidagi X va Y o‘qi shkalalarini o‘qib va qayd qilib qo‘ysangiz, namuna siljitilgandan keyin ham dastlabki kuzatuv joyiga oson qaytishingiz mumkin.

< Shkalani o‘qish >

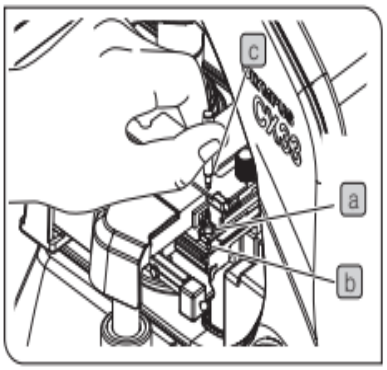
1) Ichki shkala (asosiy shkala) yordamida tashqi shkalaning (vernyer shkala) “0” nuqtasini o‘qing. **"a"**

2) Tashqi shkala chizig‘i ichki shkala chizig‘i bilan mos kelgan nuqtani tashqi shkala (vernyer) bo‘yicha o‘qing. **"b"**

Shkala birliklari: 0,1 mm.

MASLAHAT (TIP): Chapdagi rasmda ko‘rsatilgan holat 137,8 mm deb o‘qiladi.

Stolchani mahkamlash (Fixing the stage)

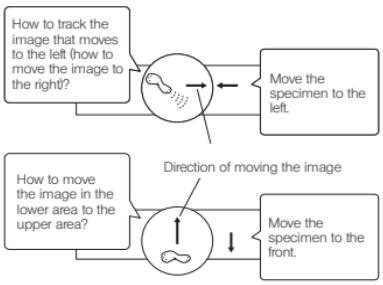


Agar namuna ushlagichdan foydalanmasdan, namunani qo‘l bilan harakatlantirib kuzatuv nuqtasini o‘zgartirmoqchi bo‘lsangiz, stolchani kutilmagan siljishlarning oldini olish uchun mahkamlash mumkin.

1) X va Y o‘qi tugmalarini harakatlantirib, stolchaning orqa o‘ng qismidagi teshik **“a”** ni vint teshigi **“b”** bilan mos keltiring.

2) Stolchani mahkamlash tugmasi **“c”** ni **“a”** teshik orqali kiriting va **“b”** vint teshigiga burab mahkamlang.

ESLATMA : Stolchani mahkamlash tugmasi mikroskop bilan birga beriladi. Uni yo‘qotib qo‘ymaslik uchun xavfsiz joyda saqlang.



Kuzatuv tasvirini qanday kuzatish mumkin

ESLATMA : Okulyar orqali kuzatilgan tasvirning yuqoriga-pastga va o‘ngga-chapga harakatlari namunaning harakatlariga teskari bo‘ladi.

4. Obyektivni tanlash (Selecting the objective)



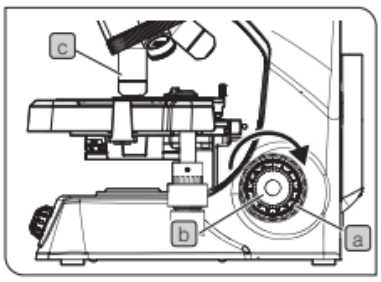
1) Aylanuvchi revolver (nosepiece) **“a”** ni ushlab, kerakli obyektiv namunaga aniq yuqoridan mos kelguncha aylantiring.

ESLATMA :

· Revolverni obyektivdan ushlab aylantirmang.

· Yuqori kattalashtiruvchi obyektiv (40X va boshqalar) bilan predmet oynasining chetini kuzatayotganda revolverni aylantirishda ehtiyot bo‘ling, chunki obyektiv namuna ushlagichga tegib ketishi mumkin.

5. Namunaga fokuslash (Focusing on the specimen)

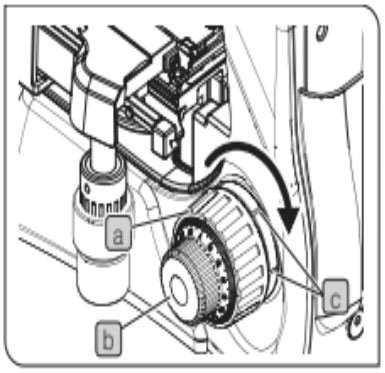


1) Mikroskopning o‘ng tomonidan qarab, makrovint(yirik fokuslash) tugmasi **“a”** ni soat strelkasi yo‘nalishida aylantirib, obyektiv **“c”** ni namunaga imkon qadar yaqinlashtiring.

2) Okulyar orqali namunani kuzatar ekan, makrovint(yirik fokuslash) tugmasi **“a”** ni strelkaga qarama-qarshi yo‘nalishda asta-sekin aylantirib, stolchani pastga tushiring.

3) Namuna ko‘rina boshlagach, aniq fokuslash tugmasi “b” ni aylantirib, tasvirni aniq fokusga keltiring.

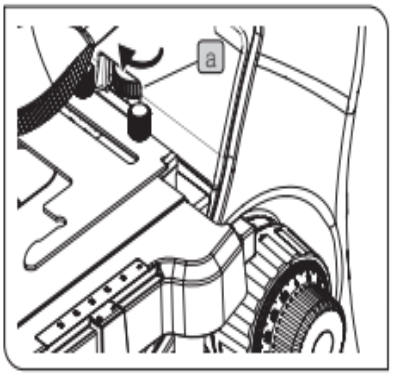
Makrovint(Yirik fokuslash) tugmasining tarangligini sozlash (Adjusting the tension of the coarse focusing knob)



Makrovint(Yirik fokuslash) tugmasining aylanish tarangligini sozlash mumkin. Buning uchun oddiy tekis uchli otvyortkani (flathead screwdriver) taranglik sozlash halqasidagi **“c”** o‘yig‘iga kiriting. Uni soat strelkasi yo‘nalishida aylantirsangiz, taranglik ortadi, qarama-qarshi yo‘nalishda aylantirsangiz, **“a”** ning tarangligi kamayadi.

MASLAHAT (TIP): Agar stolcha o‘z og‘irligi bilan pastga tushib ketsa yoki aniq fokuslash tugmasi **“b”** yordamida olingan fokus tez yo‘qolsa, demak taranglik juda bo‘sh sozlangan. Bunday holatda taranglikni oshirish uchun sozlash halqasini strelka yo‘nalishida aylantiring.

Pre-fokuslash richagidan foydalanish (Using the pre-focusing lever)



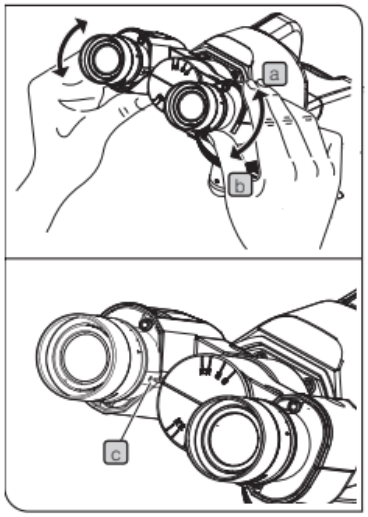
Pre-fokuslash richagi namuna va obyektiv to‘qnashib ketishining oldini olib, namunani shikastlanishdan himoya qiladi.

1) Yuqori kattalashtiruvchi obyektiv yordamida namunani fokusga keltirgandan so‘ng, pre-fokuslash richagi “a” ni strelka yo‘nalishida to‘xtaguncha aylantiring.

2) Fokuslash uchun biroz zaxira masofa qoldirish maqsadida, richag “a” ni to‘xtagan holatidan strelkaga qarama-qarshi yo‘nalishda taxminan yarim aylanishga burang.

ESLATMA: Agar pre-fokuslash richagi to‘g‘ri holatga o‘rnatilmasa, stolcha yuqoriga ko‘tarilmasligi va namuna fokusga kelmasligi mumkin. Bunday holatda richag holatini qayta sozlang.

6. Interpupillyar masofani sozlash (Adjusting the interpupillary distance)



Interpupillyar masofani sozlash — bu ikki okulyar orasidagi masofani ko‘zlaringiz orasidagi masofaga moslashtirishdir. Bu bitta aniq tasvirni ko‘rishga va kuzatuv vaqtida ko‘z charchashini kamaytirishga yordam beradi.

1) O‘ng va chap okulyarlarni gorizontal holatda tuting. Okulyarlardan qarab turib, binokulyar qismni **“a”** yoki **“b”** yo‘nalishda siljitib, o‘ng va chap ko‘rish maydonlari to‘liq birlashguncha sozlang.
Chap okulyar trubkasidagi indikator **“c”** ko‘rsatgan qiymat sizning interpupillyar masofangizni bildiradi.

MASLAHAT (TIP):

· Keyingi foydalanish uchun interpupillyar masofangizni yozib qo‘yish tavsiya etiladi.

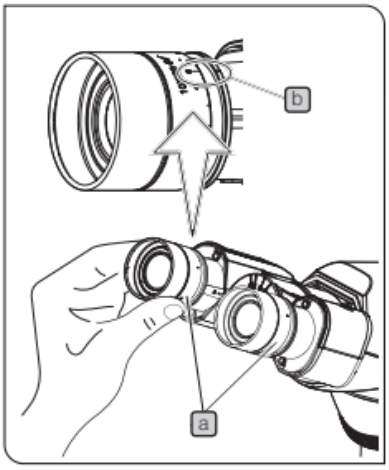
· Binokulyar qismni yuqoriga yoki pastga siljitib ham interpupillyar masofani sozlash mumkin. Kuzatuv holatingizga qarab qulay tomonni tanlang.

7. Dioptriyani sozlash (Adjusting the diopter)

Dioptriyani sozlash — bu kuzatuvchining chap va o‘ng ko‘zlari ko‘rish qobiliyatidagi farqni qoplash uchun amalga oshiriladi.

1) O‘ng va chap okulyarlardagi dioptriya sozlash halqalari “a” ni aylantirib, har ikkala shkala “0” ni indeks “b” ga moslang.

2) 10X obyektivni ish holatiga keltiring va yirik/aniq fokuslash tugmalari yordamida namunani fokusga keltiring.



3) 40X obyektivga o‘ting va yana fokuslash tugmalari yordamida namunani fokusga keltiring.

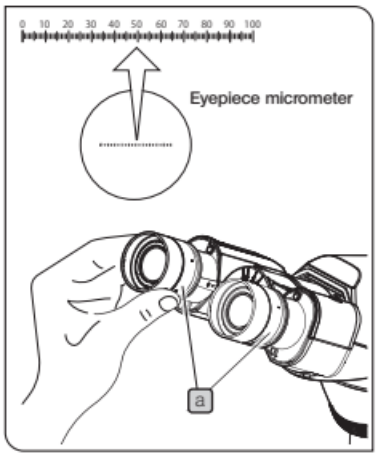
4) Yana 10X obyektivga qayting. O‘ng ko‘zingiz bilan o‘ng okulyardan qarab, dioptriya halqasi “a” ni aylantirib tasvirni aniq fokusga keltiring.
Xuddi shu tarzda chap ko‘zingiz bilan chap okulyardan qarab, dioptriya halqasini aylantirib tasvirni fokusga keltiring.

5) Yana 40X obyektivga o‘ting va fokuslash tugmalari yordamida namunani aniq fokusga keltiring.

6) 10X obyektivga qayting va o‘ng hamda chap okulyarlar orqali qarab, tasvir fokusda ekanligini tekshiring.

MASLAHAT (TIP):
Agar tasvir fokusda bo‘lmasa, 4) dan 6) bosqichlargacha qayta takrorlang.

Okulyar mikrometri o‘rnatilganda (When the eyepiece micrometer is attached to the eyepiece)

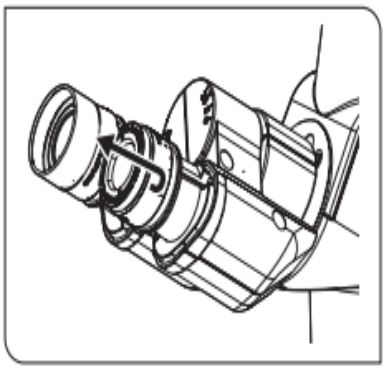


1) Okulyar mikrometri o‘rnatilgan okulyardan qarab, dioptriya sozlash halqasi “a” ni aylantirib, mikrometr shkalasi va chiziqlari aniq ko‘rinadigan qilib sozlang.

2) 10X obyektivni ish holatiga keltiring. Mikrometr o‘rnatilgan okulyardan qarab, yirik/aniq fokuslash tugmalari yordamida namunani fokusga keltiring.

3) Mikrometr o‘rnatilmagan okulyardan qarab, dioptriya sozlash halqasi “a” ni aylantirib, namunani fokusga keltiring.

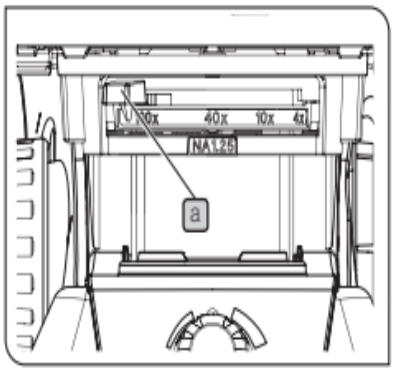
Ko‘z soyabonlaridan foydalanish (Using the eye shades)



Ko'zoynak taqqan paytda
Ko‘z soyabonlarini buklangan (pastga tushirilgan) holatda ishlating.

Ko‘zoynaksiz ishlaganda:
Buklangan ko‘z soyabonlarini strelka yo‘nalishida chiqarib (ochib) qo‘ying.
Ko‘z soyabonlari okulyar va ko‘z orasiga ortiqcha yorug‘lik kirishining oldini oladi, natijada namunani qulay kuzatish mumkin.

8. Apertura diafragmasini sozlash (Adjusting the aperture diaphragm – AS)

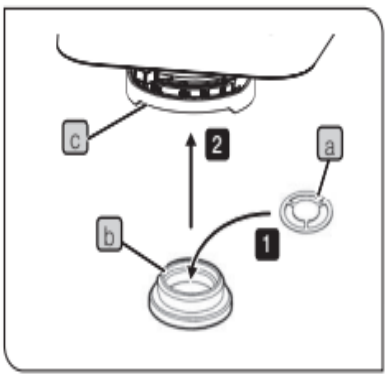


Apertura diafragmasi — bu kondensorning sonli aperturasini (numerical aperture) sozlash uchun ishlatiladigan diafragmadir. Kondensorning sonli aperturasini har bir obyektivning aperturasiga moslab sozlash orqali namunani eng yaxshi kontrast va aniqlik bilan kuzatish mumkin..

1) Apertura diafragmasi richagi “a” ni o‘ng va chap yo‘nalishda harakatlantirib, indeksni ishlatilayotgan obyektiv kattalashtirishiga mos keltiring.

ESLATMA : Qorong‘i maydon (darkfield) kuzatuvida diafragmani to‘liq oching.

9. Qorong'u maydon halqasi tirqishini mahkamlash (faqat qorong'u maydonni kuzatish uchun)

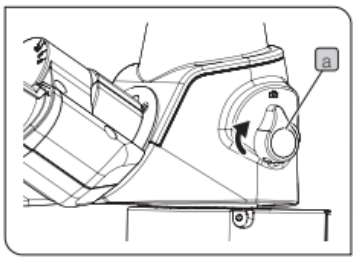


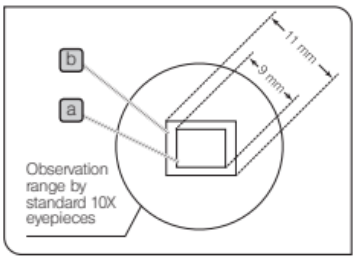
Qorong‘i maydon (darkfield) kuzatuvi uchun kondensorga darkfield halqa sliti CH2-DS (ixtiyoriy) va filtr ushlagichi CH2-FH (ixtiyoriy) o‘rnatilishi kerak.

Darkfield halqa CH2-DS “a” ni filtr ushlagichi CH2-FH “b” ichiga joylashtiring.

ESLATMA : 100X moyli immersion obyektiv bilan qorong‘i maydon kuzatuvini amalga oshirib bo‘lmaydi.

10. Kamera yordamida tasvirni olish (Acquiring the image with the camera)





Kuzatilayotgan tasvirni olish uchun trinokulyar trubkaga kamera adapteri va mikroskop uchun raqamli kamera o‘rnatiladi.

ESLATMA : Kamera adapteridan foydalanganda parfokallikni albatta sozlang. Aks holda, okulyar orqali ko‘rilgan tasvir va kamera orqali olingan tasvir bir xil fokusda bo‘lmaydi.
Parfokallikni sozlash tartibi uchun kamera adapterining qo‘llanmasiga murojaat qiling.

Trinokulyar trubkadagi yorug‘lik yo‘li tanlash tugmasi “a” ni aylantirib, kerakli holatga o‘tkazing.

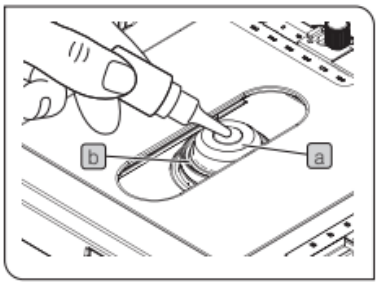
MASLAHAT : Tasvir olish maydoni kamera sensori o‘lchamiga va kamera adapterining kattalashtirish darajasiga bog‘liq.
Chapdagi rasmda standart 10X okulyar orqali kuzatish maydoni bilan kamera orqali tasvir olish maydoni taqqoslab ko‘rsatilgan.

"a" Mikroskop uchun raqamli kamera (DP22)
Tasvir sensori o'lchami (diagonal): 1/1.8 dyuym (9 mm)

"b" Mikroskop uchun raqamli kamera (DP27) Tasvir sensori o'lchami (diagonal): 2/3 dyuym (11 mm)

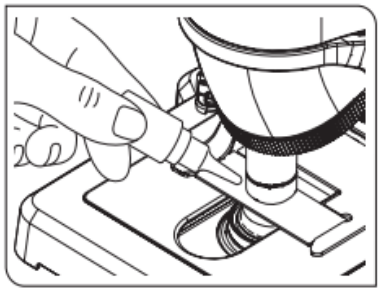
11. 100X moyli immersion obyektivdan foydalanish

1) Avval kondensor linzasi uchi “a” ga immersion moyni tomizing, so‘ng namunani joylashtiring.



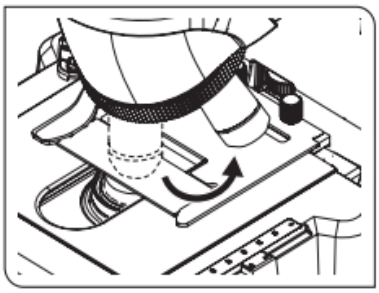
2) Past kattalashtiruvchi obyektiv yordamida namunani fokusga keltiring.

3) 100X moyli immersion obyektivni ishga kiritishdan oldin, namunadagi kuzatuv nuqtasiga immersion moyini tomizing.



4) Revolverni aylantirib, moyli immersion obyektivni yorug‘lik yo‘liga keltiring va aniq fokuslash tugmasi yordamida namunani fokusga keltiring.

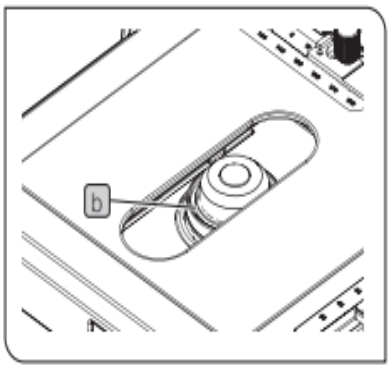
ESLATMA : Agar immersion moyida havo pufakchalari bo‘lsa, tasvir sifati yomonlashadi. Moyda pufakchalar yo‘qligiga ishonch hosil qiling.
Pufakchalarni yo‘qotish uchun revolverni biroz aylantirib, moyli obyektivni bir-ikki marta oldinga-orqaga harakatlantiring.



5) Ish tugagach, stolchani pastga tushiring, revolverni aylantiring va moyli obyektivni namunadan ajrating.

6) Immersion moyini obyektiv uchidan va kondensor linzasi uchidan tozalash qog‘ozi yoki ozgina spirt bilan namlangan mato yordamida yaxshilab artib tashlang. Xuddi shu usulda namunadagi moyni ham tozalang.

ESLATMA :



· Agar immersion moyini artib tashlamasangiz, u yopishib qolib, noto‘g‘ri kuzatuvga olib kelishi mumkin.

Moy yig‘uvchi idish “b” kondensor linzasidan oqib tushgan immersion moyini vaqtincha to‘plab turadi. Moy to‘lib ketmasligi uchun uni muntazam tozalab turing.

· Agar immersion moy “b” idishidan toshib ketsa, uni tozalash qog‘ozi yoki ozgina spirt bilan namlangan mato yordamida artib tashlang.
Agar moy yetib bo‘lmaydigan joyda bo‘lsa, pinset yordamida tozalash qog‘ozi yoki matoni ushlab, moyni artib tashlang.

V. Optik xususiyatlar atamalari lug‘ati

1. Umumiy kattalashtirish (Total magnification)

Kuzatilayotgan namuna tasvirining kattaligi okulyar kattalashtirishini obyektiv kattalashtirishiga ko‘paytirish orqali aniqlanadi. Bu qiymat umumiy kattalashtirish deb ataladi.

Misol: Okulyar (10X) × Obyektiv (40X) = 400X

2. Ajratuvchanlik (Resolution)

Ajratuvchanlik — bu linzaning bir-biriga juda yaqin joylashgan nuqtalardan hosil bo‘lgan tasvirlarni alohida ko‘rsata olish qobiliyatidir.

Ajratuvchanlik asosan obyektivning xususiyatiga bog‘liq bo‘lib, okulyarning ta’siri juda kam. Okulyarning vazifasi — obyektiv tomonidan allaqachon ajratilgan tasvirni kattalashtirishdan iborat..

3. Ko‘rish maydoni soni (Field number)

Ko‘rish maydoni soni — bu okulyar orqali ko‘riladigan tasvir diametri bo‘lib, millimetrda ifodalanadi.

Bir xil kattalashtirishda, ko‘rish maydoni soni qanchalik katta bo‘lsa, bir vaqtning o‘zida kuzatiladigan maydon shunchalik keng bo‘ladi.

Standart 10X okulyar: 20 mm

WHSZ15×-H: 16 mm

4. Haqiqiy ko‘rish maydoni (Actual field of view)

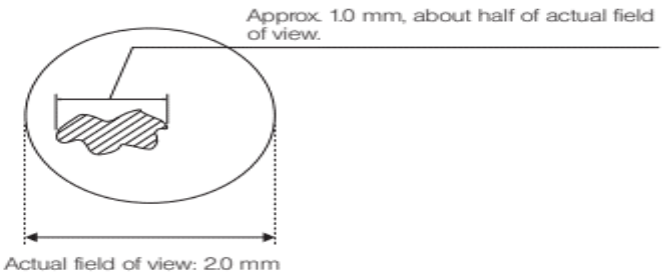
Haqiqiy ko‘rish maydoni — bu okulyar orqali amalda ko‘rinayotgan namunaning o‘lchamidir.

Bu orqali namunaning taxminiy hajmini aniqlash mumkin..

$$\text{Actual field of view} = \frac{\text{Field number}}{\text{Magnification of objective}}$$

Example : If the field number of the eyepiece is 20 and the magnification of the objective is 10X,

$$\text{Actual field of view} = \frac{20}{10} = 2.0 \text{ mm}$$



5. Apertura diafragmasi (Aperture diaphragm)

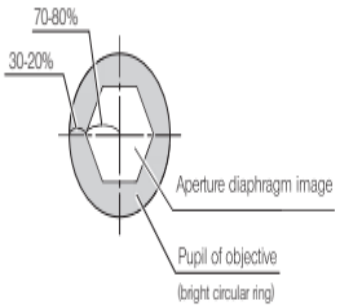
Apertura diafragmasi — bu kondensorning sonli aperturasini (numerical aperture) sozlash uchun ishlatiladigan diafragmadir.

Kondensorning sonli aperturasini obyektivning sonli aperturasiga moslab sozlash orqali namunani eng yaxshi kontrast va aniqlik bilan kuzatish mumkin. Umuman olganda, mikroskopda kuzatiladigan namunaning kontrasti past bo‘lgani sababli, kondensor aperturasini obyektiv aperturasining taxminan 70–80% ga moslab sozlash tavsiya etiladi.

Taxminiy ko‘rsatkich sifatida: okulyarsiz trubkaga qaralganda ko‘rinadigan diafragma tasviri o‘ng tomondagi rasmga o‘xshash bo‘lishi kerak..

Numerical aperture of objective and numerical aperture of condenser suitable for observation

Objective	Magnification	Numerical aperture of objective NA	Numerical aperture of condenser suitable for observation of specimen for microscope (when setting to 70% of objective)
Plan objective	4X	0.10	0.10 x 0.7 = 0.07
	10X	0.25	0.25 x 0.7 = 0.175
	20X	0.40	0.40 x 0.7 = 0.28
	40X	0.65	0.65 x 0.7 = 0.455
	100XO	1.25	1.25 x 0.7 = 0.875



VI. Muammolarni bartaraf etish

Muammo	Sabab	Chora
1. Kuzatilayotgan ko‘rish maydonining yorqinligi notekis	Trubkadagi yorug‘lik yo‘lini tanlash tugmasi to‘g‘ri holatda emas	Yorug‘lik yo‘lini tanlash tugmasini to‘xtaguncha to‘liq aylantiring.
	Obyektiv yorug‘lik yo‘liga to‘liq kiritilmagan	Revolverni (aylanuvchi bosh qismni) aylantirib, obyektiv to‘liq joylashguncha (klik ovozi chiqquncha) burang.
	Obyektiv, okulyar, kondensor, oynacha (linza) yoki namuna ifloslangan	Ularni yaxshilab tozalang.
2. Ko‘rish maydonida chang yoki dog‘lar ko‘rinmoqda	Obyektiv, okulyar, kondensor, oynacha (linza) yoki namuna ifloslangan	Ularni yaxshilab tozalang.
3. Kuzatilayotgan tasvir haddan tashqari yorqin (yaltiroq)	Apertura diafragmasi juda tor qilib qo‘yilgan	Apertura diafragmasi richagi tugmasini ishlatilayotgan obyektiv kattalashtirishiga mos holatga keltiring.
4. Kuzatilayotgan tasvir oq rangda xiralashgan yoki aniq emas	Obyektiv yorug‘lik yo‘liga to‘liq kiritilmagan	Revolverni (aylanuvchi bosh qismni) aylantirib, obyektiv to‘liq joylashguncha (klik ovozi chiqquncha) burang.
	Obyektiv, okulyar, kondensor, oynacha (linza) yoki namuna ifloslangan	Ularni yaxshilab tozalang.
	Moyli immersion obyektiv bilan immersion moyi ishlatilmagan	Immersion moyidan foydalaning.
	Immersion moyida havo pufakchalari bor	Havo pufakchalarini olib tashlang.
	Belgilangan (mos) immersion moyi ishlatilmagan	Olympus tomonidan taqdim etilgan immersion moyidan foydalaning.

Muammo	Sabab	Chora
5. Tasvirning bir tomoni xiralashgan, tasvir “oqayotgandek” ko‘rinadi.	Obyektiv yorug‘lik yo‘liga to‘liq kiritilmagan.	Revolverni aylantirib, obyektiv to‘liq joylashguncha (klik ovozi chiqquncha) burang.
	Namuna stolchaga to‘g‘ri joylashtirilmagan	Namunani stolchaga to‘g‘ri joylashtiring va uni namuna ushlagichi yordamida mahkamlang.
6. Yuqori kattalashtiruvchi obyektiv namuna fokusga kelishidan oldin unga tegib ketmoqda.	Namuna teskari joylashtirilgan	Namunani qoplovchi oyna (cover glass) tomoni yuqoriga qaratilgan holda joylashtiring.
7. Yirik fokuslash tugmasining tarangligi juda qattiq	Taranglikni sozlash halqasi haddan tashqari qattiq tortilgan	Mos (qulay) aylanishni ta’minlash uchun taranglik sozlash halqasini biroz bo‘shating
8. Namuna fokusga kelmayapti (stolcha yuqoriga ko‘tarilmayapti)	Pre-fokuslash richagi juda past holatda o‘rnatilgan	Uning pozitsiyasini ko'taring.
9. Stolcha o‘z og‘irligi bilan pastga tushmoqda yoki fokus yo‘qolmoqda (yirik fokuslash tugmasi sirg‘almoqda)	Taranglikni sozlash halqasi juda bo‘sh qo‘yilgan	Mos (qulay) taranglikni ta’minlash uchun taranglik sozlash halqasini qattiqroq qilib torting
10. Ko‘rish maydoni ikki ko‘z uchun mos kelmayapti.	Interpupillyar masofa to‘g‘ri sozlanmagan	Uni to'g'ri sozlang.
	Ikki ko‘z dioptriyasi farqi to‘g‘ri kompensatsiya qilinmagan	Uni to'g'ri tuzating.
	O‘ng va chap tomonda turli okulyarlardan foydalanilgan	O'ng va chap tomonlar uchun bir xil okulyardan foydalaning.

Muammo	Sabab	Chora
11. Past kattalashtirishdan yuqori kattalashtirishga o‘tganda obyektiv namunaga tegib ketmoqda	Namuna teskari joylashtirilgan	Namunani qoplovchi oyna (cover glass) tomoni yuqoriga qaratilgan holda joylashtiring.
	Qoplovchi oyna (cover glass) juda qalin	Qalinligi 0,17 mm bo‘lgan qoplovchi oynadan foydalaning.
12. LED yoritish ishlamayapti.	AC adapteri yoki quvvat simi uzilgan.	Elektr simini rozetkaga mahkam ulang.
13. Kattalashtirishni o‘zgartirganda namuna fokusdan chiqib ketmoqda.	Dioptriya noto‘g‘ri sozlangan.	Okulyarlarning dioptriyasini to‘g‘ri sozlang